

Two open issues of the cut-cell IBM

Rotating-wall mass source and second-order pressure gradient: analysis, proposed fixes, verification plan

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Notation as in `doc/cutCellIBM.pdf`: θ_f face and α_P cell fluid fractions, $\mathbf{S}_f$ face area vector out of P , $\mathbf{S}_w = -\sum_f \theta_f \mathbf{S}_f$ the wall vector (from the fluid into the body), $A_{\text{wall}} = |\mathbf{S}_w|$, $\mathbf{n}_w = \mathbf{S}_w/A_{\text{wall}}$, $\mathbf{x}_P$ cell centre, $\mathbf{x}_c$ fluid centroid, $\mathbf{x}'_f$ wet-face centroid, $\mathbf{u}_b(\mathbf{x}) = \mathbf{U} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c})$ the rigid body velocity. All numbers below are computed with the geometry algorithm of the library (bisection, 32×32 sub-sampling, absorption below $\alpha_{\min} = 0.02$) for a cylinder of radius 1; the scripts are next to this file.

	branches	problem	proposed fix
1. rotating-wall mass source	<code>movingBody</code> , <code>scalar</code> ; reduced on <code>scalar_secondOrder</code> , <code>fouling</code>	$S_P = \mathbf{u}_b(\mathbf{x}_P^e) \cdot \mathbf{S}_w$ is not zero for a wall moving tangentially: spurious local sources of order $ \boldsymbol{\omega} hA_{\text{wall}}$; with the centroid point also $\sum_P S_P \neq 0$	$S_P = \mathbf{U} \cdot \mathbf{S}_w + S_P^{\text{rot}}$, the rotational part from the geometric wet faces; exact flux through the discrete wall, $\sum_P S_P = 0$ for any motion and mesh
2. second-order pressure gradient	<code>scalar_secondOrder</code> , <code>fouling</code>	face values sit at the centroid midpoint, $\theta_f = 1$ faces are not corrected, the wall value is at the normal foot of the centroid: errors of 17% (median) in the cut cells for a linear p	face correction relative to the centroid midpoint on every face of a cut cell; wall pressure integrated over the wall pieces: exact for linear p

1 Issue 1: mass source of a rotating wall

1.1 What the code does

For a moving body the continuity equation of a wall cell reads $\sum_f \phi_f = -S_P$ and the old fluid fraction is recovered from the same source, $\alpha_P^n V_P = \alpha_P^{n+1} V_P - \Delta t S_P$. The source is evaluated with the body velocity at one point per cell,

$$S_P = \mathbf{u}_b(\mathbf{x}_P^e) \cdot \mathbf{S}_w, \quad \mathbf{x}_P^e = \begin{cases} \mathbf{x}_P + d_{\text{wall}} \mathbf{n}_w & \text{movingBody, scalar,} \\ \mathbf{x}_{c,P} + d_{\text{wall}} \mathbf{n}_w & \text{scalar_secondOrder, fouling.} \end{cases} \quad (1)$$

For a translation $\mathbf{u}_b$ is uniform and (1) is exact. For a rotation it is not.

1.2 Why a point evaluation cannot work

Lemma 1. For a rigid velocity and any s , $\mathbf{u}_b(\mathbf{x} + s\mathbf{n}_w) \cdot \mathbf{n}_w = \mathbf{u}_b(\mathbf{x}) \cdot \mathbf{n}_w$.

Proof. $(\boldsymbol{\omega} \times s\mathbf{n}_w) \cdot \mathbf{n}_w = 0$. □

Lemma 2. For a planar surface Γ with area vector $\mathbf{S}_\Gamma$ and centroid $\mathbf{x}_\Gamma$, $\int_\Gamma \mathbf{u}_b \cdot \mathbf{n} dS = \mathbf{u}_b(\mathbf{x}_\Gamma) \cdot \mathbf{S}_\Gamma$.

Proof. $\mathbf{u}_b$ is linear in $\mathbf{x}$ and $\mathbf{n}$ is constant on Γ . $\square$

In two dimensions the wall of a cut cell is the chord between the two edge crossings, and $\mathbf{S}_w$ is normal to it. For a body of revolution rotating about its axis, the chord midpoint $\mathbf{x}_m$ lies on the line through the centre along $\mathbf{S}_w$, so $\mathbf{u}_b(\mathbf{x}_m) \cdot \mathbf{S}_w = 0$: the exact flux through the discrete wall vanishes, as it must, since the geometry does not change. A point evaluation gives instead

$$S_P = \mathbf{u}_b(\mathbf{x}_P^e) \cdot \mathbf{S}_w = (\boldsymbol{\omega} \times (\mathbf{x}_P^e - \mathbf{x}_m)) \cdot \mathbf{S}_w = \boldsymbol{\omega} \cdot ((\mathbf{x}_P^e - \mathbf{x}_m) \times \mathbf{S}_w), \quad (2)$$

which depends only on the offset of $\mathbf{x}_P^e$ across the normal (Lemma 1). The cell centre is offset by $O(h)$, so $|S_P| = O(|\boldsymbol{\omega}|hA_{\text{wall}})$ with alternating sign around the body. Moving the point along $\mathbf{n}_w$, as on `movingBody` and `scalar`, changes nothing: Fig. 1b shows the two forms coincide to round-off. The centroid of `scalar_secondOrder` is not on the line either, and the spurious source is of the same size (Fig. 1b); in the Taylor–Couette case it happened to be smaller than with the cell centre.

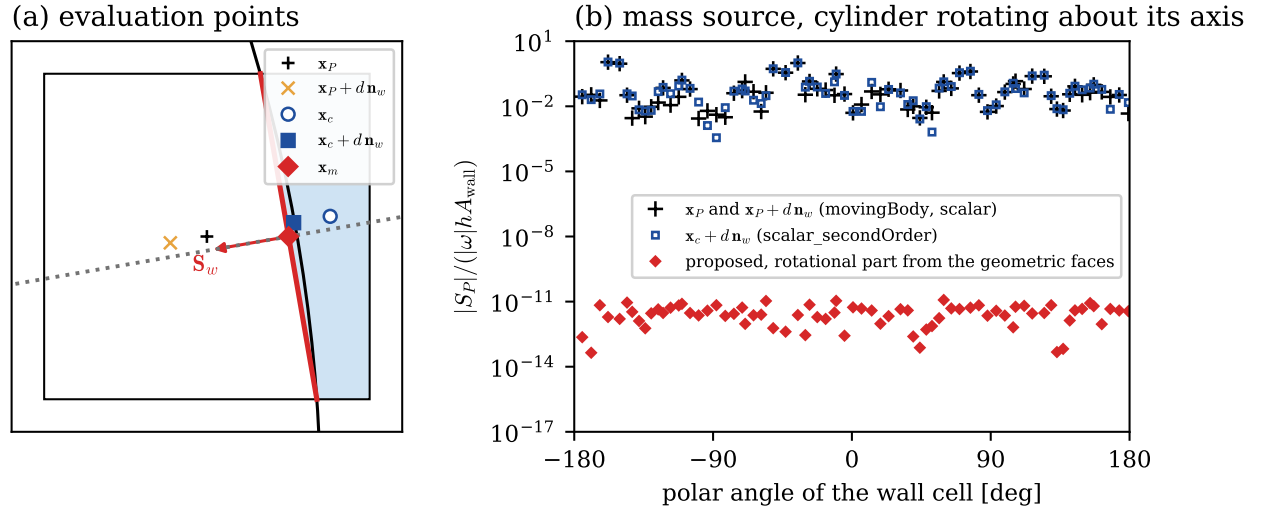


Figure 1: (a) A cut cell of a cylinder rotating about its axis (dotted: the line through the centre along $\mathbf{S}_w$). $\mathbf{x}_P$ and $\mathbf{x}_P + d\mathbf{n}_w$ lie on a common normal and give the same source; only points on the dotted line, such as the chord midpoint $\mathbf{x}_m$, give zero. (b) Source of every wall cell for $\boldsymbol{\omega} = 1$, $h = 0.1$, normalised with $|\boldsymbol{\omega}|hA_{\text{wall}}$.

Consequences. (i) A body rotating in place has static geometry, yet the solver assigns $\alpha_P^n \neq \alpha_P^{n+1}$ and a non-zero divergence to its wall cells; the resulting local sources checkerboard the cut-cell pressure (errors of ± 0.4 in the Taylor–Couette case before the centroid evaluation, for a pressure range of 0.08). (ii) Compatibility. Since $\sum_P \mathbf{S}_{w,P} = \mathbf{0}$ for a closed body,

$$\sum_P S_P = \boldsymbol{\omega} \cdot \sum_P \mathbf{x}_P^e \times \mathbf{S}_{w,P} = \boldsymbol{\omega} \cdot \sum_f \theta_f (\mathbf{x}_N^e - \mathbf{x}_O^e) \times \mathbf{S}_f. \quad (3)$$

With $\mathbf{x}^e = \mathbf{x}_P + d\mathbf{n}_w$ this is the cell-centre form, $d_f \times \mathbf{S}_f$, which vanishes on orthogonal meshes. With the centroid it does not vanish: for the cylinder of Fig. 1 $\sum_P S_P = -1.2 \times 10^{-3}$, against individual sources of 10^{-2} . By the reference lemma of `cutCell1IBM.pdf` (Lemma 1) this defect is deposited at the pressure reference cell as a point source. The Taylor–Couette case logged 10^{-17} only because its geometry is mirror-symmetric.

1.3 Proposed fix

Split the motion into the translation of the reference point $\mathbf{c}$ and the rotation about it. The translational part is kept as it is, because it is what makes free-stream preservation exact (it includes the faces closed by sliver absorption, which carry the swept volume of the absorbed cell). The rotational part is computed as a flux through the geometric wall. Since $\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c})$ is divergence-free and linear, the divergence theorem over the geometric fluid polygon Ω_P^{geo} of cell P (bounded by the wet faces before any post-processing and by the chord) gives exactly

$$S_P^{\text{geo}} \equiv \int_{\text{chord}} (\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c})) \cdot \mathbf{n} \, dS = - \sum_{f \in P} \theta_f^{\text{geo}} (\boldsymbol{\omega} \times (\mathbf{x}'_f^{\text{geo}} - \mathbf{c})) \cdot \mathbf{S}_f, \quad (4)$$

with θ_f^{geo} and $\mathbf{x}'_f^{\text{geo}}$ the fraction and wet-face centroid of the face *before* absorption and synchronisation (a face value of a linear field over a planar face is the value at its centroid). Cells removed by the post-processing (absorbed slivers, and cells with $\alpha = 0$ from the sampling but wet faces) form the set $\mathcal{R}$; their S^{geo} is handed to their fluid face neighbours in proportion to the shared geometric wet area. The proposed source is

$$S_P = \mathbf{U} \cdot \mathbf{S}_{w,P} + S_P^{\text{geo}} + \sum_{N \in \mathcal{R}, N \sim P} w_{NP} S_N^{\text{geo}}, \quad w_{NP} = \frac{\theta_{f_{NP}}^{\text{geo}} |\mathbf{S}_{f_{NP}}|}{\sum_{f \in N, \text{ other side fluid}} \theta_f^{\text{geo}} |\mathbf{S}_f|} \quad (5)$$

Proposition 1. *The source (5) has the following properties.*

1. Translation unchanged. For $\boldsymbol{\omega} = \mathbf{0}$ it is the current source; free-stream preservation (Proposition 7 of *cutCellIBM.pdf*) is unaffected.
2. Exact flux through the discrete wall. Outside the neighbourhood of $\mathcal{R}$ the rotational part is the exact flux of the rotation through the chord(s) of the cell. For a body of revolution rotating about its axis it vanishes in two dimensions (to the bisection tolerance), so a body rotating in place has $S_P = 0$ and $\alpha_P^n = \alpha_P^{n+1}$.
3. Compatibility for every rigid motion and every mesh. $\sum_P S_P = \mathbf{U} \cdot \sum_P \mathbf{S}_{w,P} - \sum_{\text{all cells}} \sum_f \theta_f^{\text{geo}} (\boldsymbol{\omega} \times (\mathbf{x}'_f^{\text{geo}} - \mathbf{c})) \cdot \mathbf{S}_f = 0$: the handover conserves $\sum S^{\text{geo}}$, and every internal face appears twice with the same θ_f^{geo} and $\mathbf{x}'_f^{\text{geo}}$ and opposite $\mathbf{S}_f$. On periodic faces the two copies are mapped to the same minimum image about $\mathbf{c}$ and cancel as well.

Numerically (Fig. 1b, $h = 0.1$, 7 cells in $\mathcal{R}$): $\max_P |S_P| = 7 \times 10^{-14}$ for the rotation about the axis (against 1.1×10^{-2} today), and $\sum_P S_P = 6 \times 10^{-17}$ also for an eccentric rotation ($\mathbf{c} = (0.3, -0.2)$), where the centroid form gives -1.2×10^{-3} .

Remark 1. In three dimensions the chord becomes the wall polygon of the cut cell; (4) is then the flux through the polyhedral approximation of the surface, which is not exactly zero for a sphere rotating about a diameter but converges to zero with the geometry. Properties 1 and 3 hold unchanged.

Remark 2. The point $\mathbf{x}_P^e$ is still needed for the *tangential* wall velocity of the no-slip traction $\nu A_{\text{wall}}(\mathbf{u}_P - \mathbf{u}_b)/d$. There the current choices are appropriate: the value at the foot of the normal from the location of the cell value, $\mathbf{x}_P + d\mathbf{n}_w$ (first order) or $\mathbf{x}_c + d_c\mathbf{n}_w$ (second order), is what the two-point wall gradient differences against. Only the mass source changes.

Implementation.

- **faceFractions**: keep θ_f^{geo} and the wet-face centroid of every face as computed, before absorption and synchronisation (**scalarSecondOrder** already computes the wet-face offset; **movingBody/scalar** need the few lines of the fan-triangulated centroid).

- **bodyMotion/ibmBody**: accessors for $\mathbf{U}$, $\boldsymbol{\omega}$ and the current centre $\mathbf{c}(t)$, and the minimum-image map already used for the level set.
- **calcMovingTerms**: $S_P = \mathbf{U}_{b(P)} \cdot \mathbf{S}_w$ as now, plus a loop over the faces for (4) and the handover of $\mathcal{R}$ (across processor faces with **syncTools**). α^n follows from S_P as now. Cost: one face loop per step.
- Cells shared by two bodies: use the body of the cell, as now; faces between cells of two different moving bodies are the one remaining case where $\sum_P S_P$ is not exact.

Verification.

1. Cylinder rotating in place (**motion {type none; angularVelocity (0 0 1);}**): $\max_P |S_P|$ at round-off, $\alpha^n = \alpha^{n+1}$, no checkerboard in the cut-cell pressure.
2. Taylor–Couette (**tutorials/order/taylorCouette**) on all moving branches, first- and second-order: velocity and pressure errors against the current ones.
3. Eccentric rotation or a rotating ellipse: $\sum_P S_P$ at round-off.
4. Regression: free stream, oscillating and translating cylinder must be bit-identical (pure translation).

2 Issue 2: second-order pressure gradient

2.1 What the code does

With **secondOrder true** the volume-integrated gradient is

$$\nabla_V p|_P = \sum_f \theta_f \left[\bar{p}_f + [0 < \theta_f < 1] \mathbf{g}_f \cdot (\mathbf{x}'_f - \mathbf{x}_f) \right] \mathbf{S}_f + [p_P + d \mathbf{g}_P \cdot \mathbf{n}_w] \mathbf{S}_w, \quad (6)$$

with $\bar{p}_f = w_f p_O + (1 - w_f) p_N$ the linear interpolate, $\mathbf{g}$ the least-squares gradient on the centroids ($\mathbf{g}_f$ its face average) and d the centroid wall distance. The comment in the code and the methods document before its revision stated that this is exact for a linear pressure.

2.2 Why it is not exact

Take $p = p_0 + \mathbf{G} \cdot \mathbf{x}$ with cell values at the centroids, $p_P = p(\mathbf{x}_{c,P})$; the least-squares gradient is then exact, $\mathbf{g} = \mathbf{G}$. The exact integral over the geometric fluid polygon is $\oint p \mathbf{n} \, dS = V_P^{\text{geo}} \mathbf{G}$. Three terms of (6) miss it:

1. The linear interpolate is the value at the weighted midpoint of the two *centroids*, $\bar{p}_f = p(\mathbf{m}_f)$, $\mathbf{m}_f = w_f \mathbf{x}_{c,O} + (1 - w_f) \mathbf{x}_{c,N}$, not at the face centre. The correction $\mathbf{G} \cdot (\mathbf{x}'_f - \mathbf{x}_f)$ therefore leaves the error $\mathbf{G} \cdot (\mathbf{m}_f - \mathbf{x}_f)$, i.e. $\frac{1}{2} \mathbf{G} \cdot (\mathbf{o}_O + \mathbf{o}_N)$ on a uniform mesh, on every partially wet face.
2. Faces with $\theta_f = 1$ next to a cut cell are not corrected at all, although $\mathbf{m}_f \neq \mathbf{x}_f$ there.
3. The wall value is taken at $\mathbf{x}_c + d \mathbf{n}_w$, the normal foot of the centroid, while the exact integral over a planar wall is the value at the wall centroid $\mathbf{x}_m$ (Lemma 2 with p for $\mathbf{u}_b$). The tangential offset gives $\mathbf{G} \cdot (\mathbf{x}_c + d \mathbf{n}_w - \mathbf{x}_m) \mathbf{S}_w$; for a cell next to an absorbed sliver or a staircase face the wall is not even planar.

Fig. 2b shows the result: relative errors of 0.17 (median) and up to 5.7 in the cut cells, not better than the first-order form (0.28 and 6.5).

2.3 Proposed fix

$$\bar{p}_f \rightarrow \bar{p}_f + \mathbf{g}_f \cdot (\mathbf{x}'_f - \mathbf{m}_f) \quad \text{on every open face of a cell with } \mathbf{o} \neq \mathbf{0} \text{ or } \theta_f < 1, \quad (7)$$

$$p_P \mathbf{S}_w \rightarrow \sum_k [p_P + \mathbf{g}_P \cdot (\mathbf{x}_k - \mathbf{x}_{c,P})] \mathbf{S}_k, \quad (8)$$

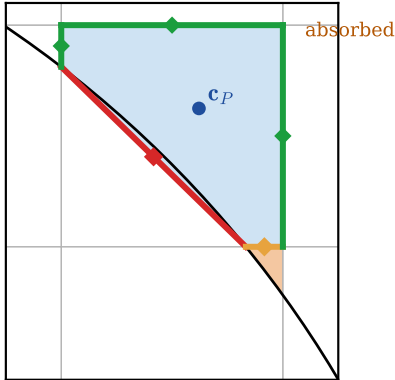
where the wall of the cell is split into pieces k with area vectors $\mathbf{S}_k$ and centroids $\mathbf{x}_k$ (Fig. 2a): the chord(s) of the geometric cut (midpoint), and every geometrically wet part of a face that the post-processing closed (area vector $\theta_f^{\text{geo}} \mathbf{S}_f$, centroid $\mathbf{x}_f^{\text{geo}}$). By the closure of the geometric polygon, $\sum_k \mathbf{S}_k = \mathbf{S}_w$. On faces between two full cells of an orthogonal mesh $\mathbf{m}_f = \mathbf{x}'_f = \mathbf{x}_f$ and (7) is zero, so the uncut part of the domain is untouched.

Proposition 2. *If the least-squares gradient is exact for linear fields, the gradient with (7) and (8) satisfies $\nabla_V p|_P = V_P^{\text{geo}} \mathbf{G}$ for every linear p and every wall cell. It vanishes for a uniform p , conserves momentum (the face corrections are antisymmetric, since $\mathbf{x}'_f$ and $\mathbf{m}_f$ are shared by both cells), and the pressure force $\mathbf{F}_p = \sum_P \sum_k p_k \mathbf{S}_k$ keeps the momentum-sum identity.*

Proof. Each face term equals $p(\mathbf{x}'_f) \theta_f |\mathbf{S}_f| \mathbf{n}_f$, the exact integral over the wet face; each wall term equals $p(\mathbf{x}_k) \mathbf{S}_k$, the exact integral over the planar piece k . Together the wet faces and the pieces are the boundary of Ω_P^{geo} , and the divergence theorem gives $V_P^{\text{geo}} \mathbf{G}$. For uniform p the sum is $p(\sum_f \theta_f \mathbf{S}_f + \sum_k \mathbf{S}_k) = \mathbf{0}$. $\square$

Numerically: over 880 wall cells (12 mesh offsets) the relative error is below 4×10^{-13} (Fig. 2b). The wall distance d is no longer needed in the gradient.

(a) faces and wall pieces



(b) linear pressure, all wall cells, 12 offsets

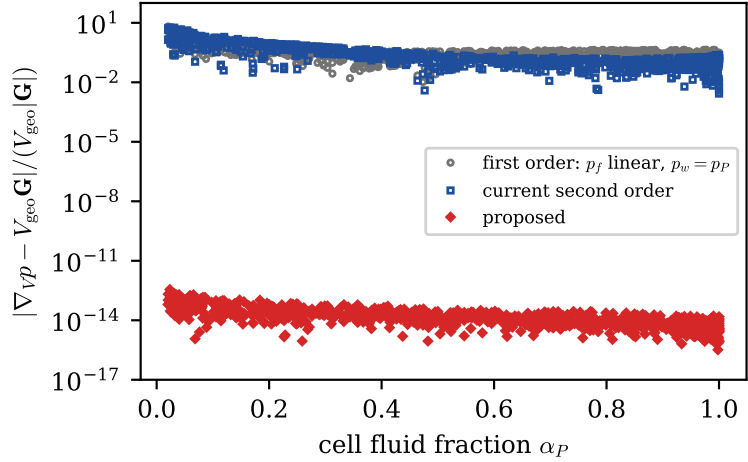


Figure 2: (a) A cell next to an absorbed sliver: open wet faces with their centroids (green), the chord with its midpoint (red) and the wet part of the face closed by the absorption (orange); the last two are the wall pieces of (8). (b) Relative error of the cut-cell gradient of a linear pressure against the exact integral over the geometric polygon.

Remark 3. In three dimensions the chord is the wall polygon formed by the crossings of the cell's cut edges (each cut face contributes one segment); its centroid by fan triangulation makes (8) exact for planar cuts and second-order accurate otherwise.

Remark 4 (What the fix does not do). It makes the gradient consistent; it does not change the conditioning of the cut-cell pressure rows (the mobility of a cut cell still scales with its fluid fraction, Section 8.6 of `cutCell1IBM.pdf`). How much of the non-convergence of the cut-cell pressure in the Taylor–Couette case is due to the gradient and how much to the conditioning is exactly what the re-run will tell; cell merging remains the remedy for the latter.

Remark 5 (Hydrostatic balance). A fluid at rest under a uniform body force $\mathbf{g}$ in a closed domain has $p = \mathbf{g} \cdot \mathbf{x}$. With the fix the pressure force of a cut cell is $V_P^{\text{geo}} \mathbf{g}$, the body force $\alpha_P V_P \mathbf{g}$; the residual $(\alpha_P V_P - V_P^{\text{geo}}) \mathbf{g}$ is the sub-sampling error of α ($\sim 10^{-3}$). Weighting the body force with the polygon volume (or computing α from the polygon, exact in two dimensions) would make the scheme exactly well balanced.

Implementation.

- `calcGeometry` (second-order path): store per wall cell the list of wall pieces (chords from the cut edges of the cell, closed geometric wet parts from the faces with $\theta_f = 0 < \theta_f^{\text{geo}}$) and $\mathbf{m}_f$ per face from `mesh.weights()` and the centroid offsets.
- `grad()`: replace the face correction by (7) on every open face touching a cell with a non-zero offset, and the wall term by (8); `forces()` uses the same pieces.

Verification.

1. `ibmOrderTest`: new problems for the gradient operator alone — linear p (error at the sub-sampling level of V^{geo} vs αV) and quadratic p (second order in the cut cells).
2. Hydrostatic box with an immersed body: spurious velocity against the current scheme.
3. Taylor–Couette pressure errors and the drag Richardson study, with the components.

3 Suggested order

1. Issue 1 on `movingBody` (about one day with tests), merged into `scalar`, `scalar.secondOrder` (where it replaces the centroid evaluation of the source) and `fouling`.
2. Issue 2 on `scalar.secondOrder` (one to two days, three-dimensional wall polygons included), merged into `fouling`; then re-run Taylor–Couette and the drag study with both fixes, and update `doc/cutCellIBM.tex` and `doc/secondOrderStudy.md`.